Further Investigation of Wall-Specific Effects on Conjugate Heat Transfer in Channel with Fins

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Introduction

Following the work of Webb and Ramadhyani [1], we use Nalu, a second-order, unstructured, multiphysics flow solver from Sandia Labs to perform conjugate heat transfer (CHT) simulations of fluid flow through a ribbed channel.

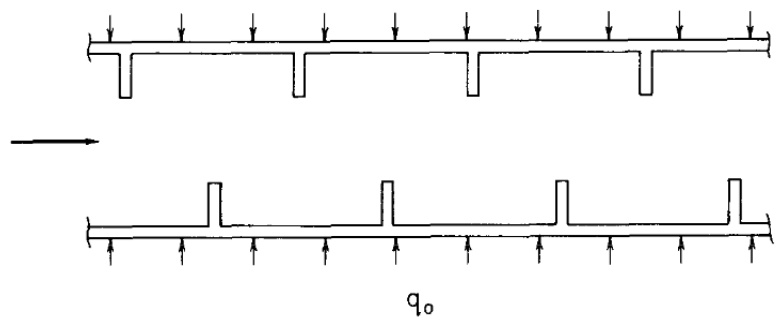


Figure 1: Ribbed channel geometry [1]

Webb and Ramadhyani [1] analyzed fluid flow and heat transfer characteristics for a constant property fluid flowing laminarly through a parallel plate channel with staggered transverse ribs and constant heat flux along both walls. After some entry length, we can say that the flow becomes fully developed and, therefore, periodic. Computations in the paper were carried out in the fully developed regime for various values of Reynolds number. They additionally varied geometrical parameters S/D and L/D (see Figure 2). The parameters W/D and B/D were held constant.

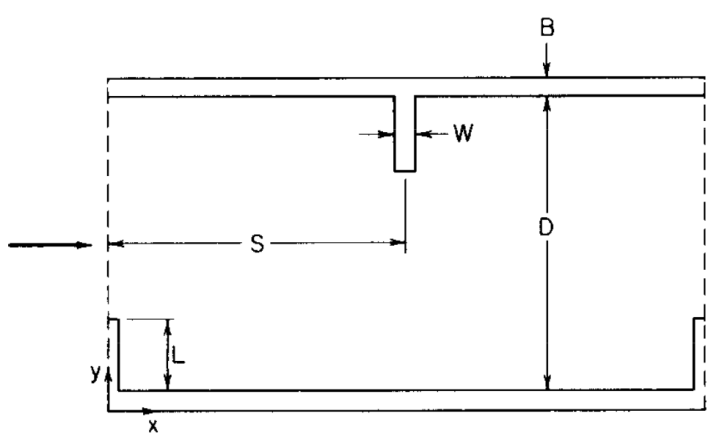


Figure 2: Channel geometry parameters and computational domain [1]

Introduction (cont.)

We seek to reproduce the results from the paper for the case of Re = 340, Pr = 0.7, $S/D = 1.0$, $L/D = 0.125$, $W/D = 0.05$, and $B/D = 0.05$. We will further extend the problem in two ways. The authors used a fixed thermal conductivity ratio $k_{\text{solid}}/k_{\text{fluid}} = 200$ and they only studied the case of $B/D = 0.05$. We will vary both these values to further investigate wall-specific effects on the conjugate heat transfer.

Governing Equations

There are two domains to consider in a conjugate heat transfer problem: the solid domain and the fluid domain. The distinguishing feature of a conjugate heat transfer problem from a traditional single-phase fluids problem is the coupled boundary condition at the interface between the fluid and the solid. In the fluid domain, the continuity, momentum, and heat equations are solved:

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} \tag{2}$$

$$u_j \frac{\partial T}{\partial x_j} = \alpha \frac{\partial^2 T}{\partial x_j^2} \tag{3}$$

In the solid domain, the heat conduction equation is solved.

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x_j^2} \tag{4}$$

At the interface between the fluid and solid, we impose the no-slip condition on our velocity field. For the thermal field, we match heat flux and temperature.

$$u = v = 0 \tag{5a}$$

$$T_{\text{solid}} = T_{\text{fluid}} \tag{5b}$$

$$k_{\text{solid}} \frac{\partial T_{\text{solid}}}{\partial n} = k_{\text{fluid}} \frac{\partial T_{\text{fluid}}}{\partial n} \tag{5c}$$

At the top and bottom walls of the solid, a uniform heat flux is applied

$$-k_{\text{solid}} \frac{\partial T}{\partial y}\bigg|_{\text{bottom wall}} = k_{\text{solid}} \frac{\partial T}{\partial y}\bigg|_{\text{top wall}} = q_0 \tag{6}$$

Periodic boundary conditions on the velocity field are imposed at the left and right boundary.

$$u(0, y) = u(2S, y) \tag{7a}$$

$$v(0, y) = v(2S, y) \tag{7b}$$

Governing Equations (cont.)

For the temperature field, Webb and Ramadhyani [1] impose a quasi-periodic boundary condition in which the global temperature rise in the channel is subtracted before the right boundary temperature is applied again to the left boundary. That is,

$$\sigma = \frac{2q_0}{\rho \overline{u} c_p D} \tag{8}$$

$$T(0, y) = T(2S, y) - \sigma \cdot 2S \tag{9}$$

Numerics

Nalu solves continuity, momentum, and enthalpy in the fluid domain in finite-volume form. For a laminar flow, the governing equations simplify to the ones shown in (10), (11), and (12).

$$\int \frac{\partial \rho}{\partial t} dV + \int \rho u_i n_i dS = 0 \tag{10}$$

$$\int \frac{\partial \rho u_i}{\partial t} dV + \int \rho u_i u_j n_j dS + \int \ P n_i dS = \int 2\mu \left(S_{ij} - \frac{1}{3} S_{kk} \delta_{ij} \right) n_j dS \tag{11}$$

$$\int \frac{\partial \rho h}{\partial t} dV + \int \rho h u_j n_j dS = \int \left(\frac{\partial P}{\partial t} + u_j \frac{\partial P}{\partial x_j} \right) dV + \int \tau_{ij} \frac{\partial u_i}{\partial x_j} dV \tag{12}$$

For the non-isothermal response that occurs in the solid, the single material heat conduction equation is solved.

$$\int \rho C_p \frac{\partial T}{\partial t} dV + \int q_j n_j dS = \int \rho C_p \frac{\partial T}{\partial t} dV - \int k \frac{\partial T}{\partial x_j} n_j dS = 0 \tag{13}$$

The interface boundary condition between the fluid and the solid is

$$q_j^{\text{solid}} n_j = q_j^{\text{fluid}} n_j = h(T - T_\infty) \tag{14}$$

Numerics (cont.)

The quasi-periodic boundary condition proposed in Webb and Ramadhyani [1] is not simple to implement in Nalu. We have instead chosen to alter the boundary condition on the top and bottom walls such that the channel is heated at the top wall and cooled at the bottom wall. That is,

$$-k_{\text{solid}} \frac{\partial T}{\partial y}\bigg|_{\text{bottom wall}} = -q_0 \tag{15a}$$

$$k_{\text{solid}} \frac{\partial T}{\partial y}\bigg|_{\text{top wall}} = q_0 \tag{15b}$$

Thus, there is no global temperature rise (the heat fluxes at the boundaries are of equal magnitude) and we can apply a regular periodic boundary condition on the thermal field:

$$T(0, y) = T(2S, y) \tag{16}$$

Webb and Ramadhyani [1] do not give the value of the pressure source term they use for their simulations. Since we do not have a simple channel and expect that the presence of ribs will introduce recirculation zones, this term is not easily calculated through analytical means. We tried multiple source terms to achieve our target Reynolds number value of 340 (Figure 3). As might be expected, the relationship between Reynolds number and the value of the source term is still (roughly) linear. Note that Reynolds number is defined using the average value of the x velocity in the channel and the hydraulic diameter $D_H = 2D$:

$$\text{Re} = \frac{\rho \overline{u} D_H}{\mu} \tag{17}$$

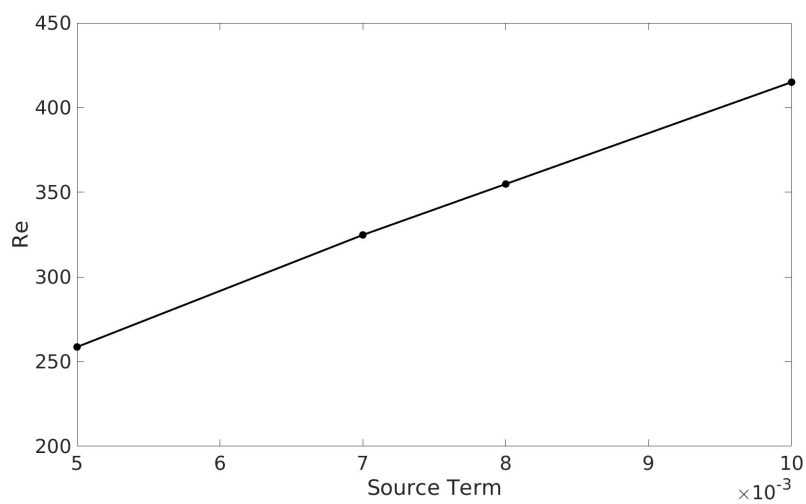


Figure 3: Reynolds number based on momentum source term value

Numerics (cont.)

Shown in Figure 4 and Figure 5 are the two meshes used in the simulations. The fluid domain in each mesh contains approximately 30,000 cells. For the case of $B/D = 0.05$, the top and bottom walls are approximately 1000 cells. For the case of $B/D = 0.10$, the top and bottom walls are approximately 2000 cells. The Pointwise TREX algorithm is used to apply targeted mesh refinement on the channel walls in order to aid in the capturing of the expected recirculation zones behind the ribs. Quadrilateral elements are used for the near-wall mesh refinement and triangular elements are used elsewhere.

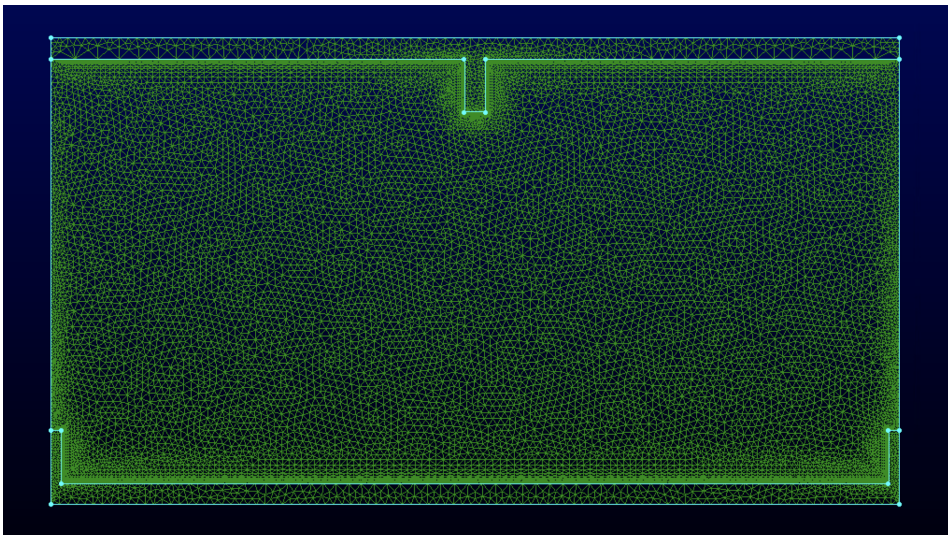


Figure 4: Mesh for the $B/D = 0.05$ geometry

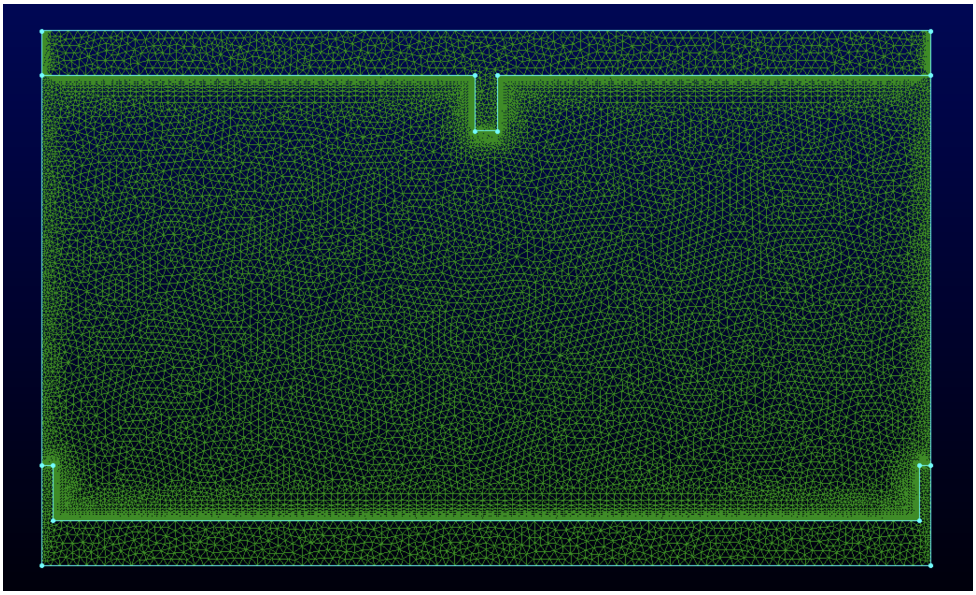


Figure 5: Mesh for the $B/D = 0.10$ geometry

Results

We first validate the Nalu simulations by comparing the velocity field from our $B/D = 0.05$ case (Figure 7) to the results in Webb and Ramadhyani (Figure 6). We note that, qualitatively, we have captured the correct velocity field. The recirculation zones behind the ribs are relatively small and the perturbation to the freestream in the y-direction is limited compared to higher Re and larger L/D cases.

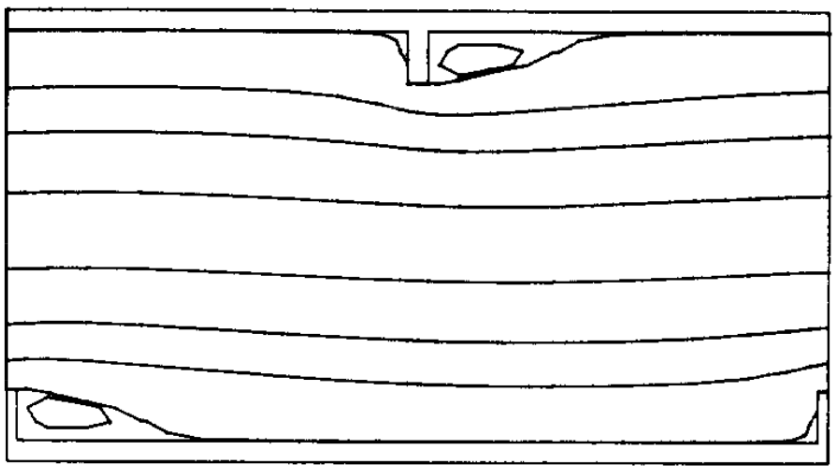


Figure 6: Streamlines of velocity field from Webb and Ramadhyani [1]

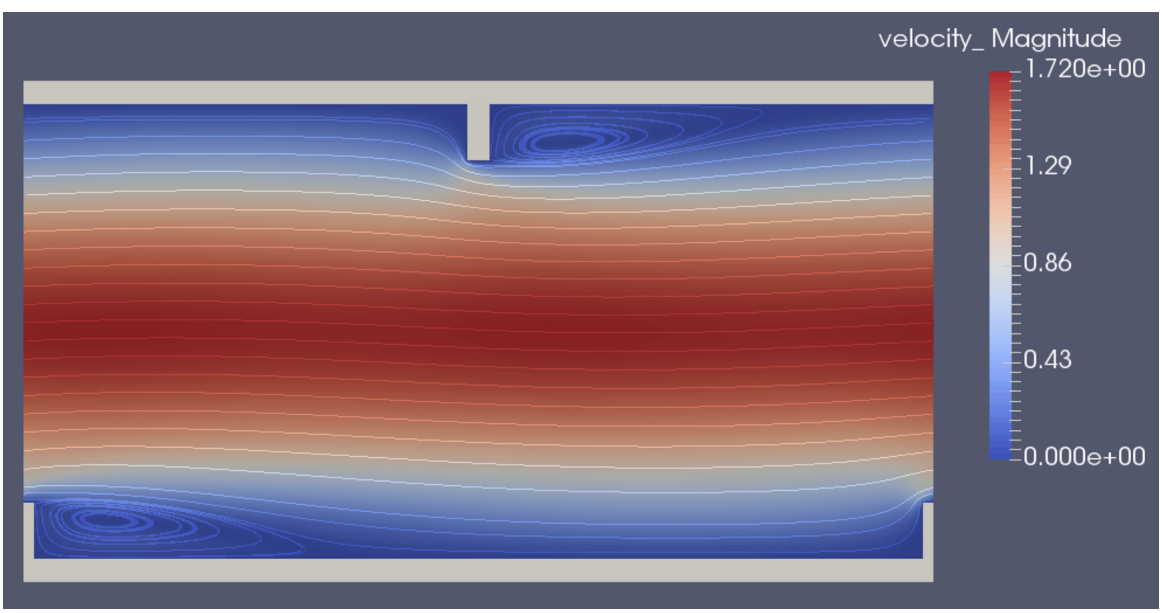


Figure 7: Streamlines of velocity field from $B/D = 0.05$ geometry

Results (cont.)

The corresponding temperature field contour is shown in Figure 8. While Webb and Ramadhyani do not provide temperature contours in their paper, this field looks reasonable. The thermal boundary layer is well-developed and follows a slight curve like the velocity streamlines. We observe a "thermal wake" behind the ribs.

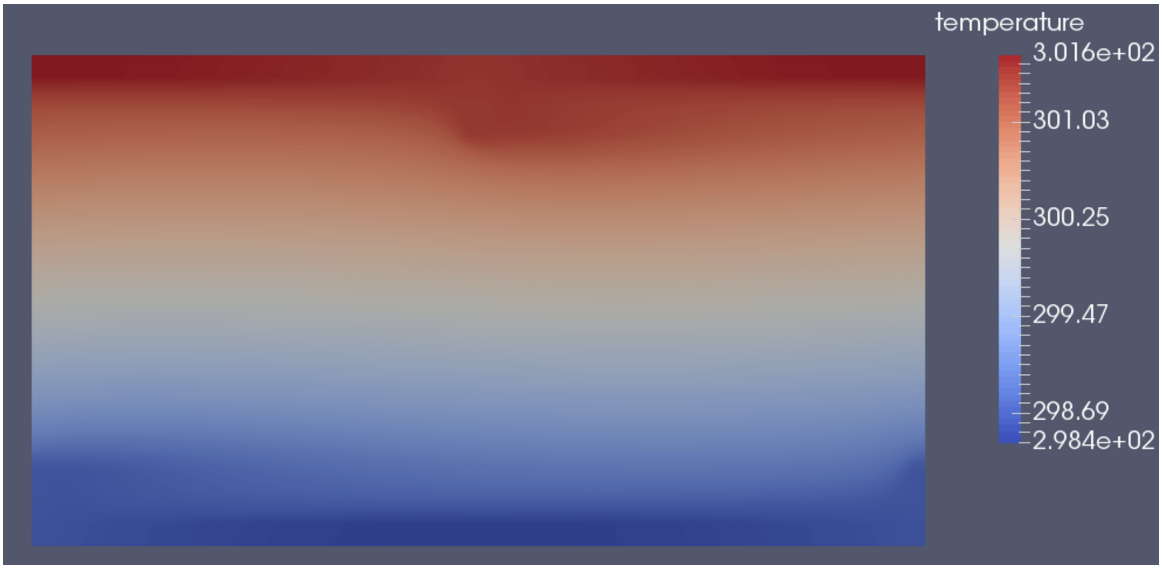


Figure 8: Temperature field (both solid and fluid domain) from $B/D = 0.05$ geometry

We further validate our simulations by comparing local heat flux values at the solid-fluid interface. Figure 9 displays this data from Webb and Ramadhyani. The values shown at the location of the rib is the heat transfer through the rib base. Note that this figure is for Re = 940 and $L/D = 0.25$ whereas we are studying the case where Re = 340 and $L/D = 0.125$.

Results (cont.)

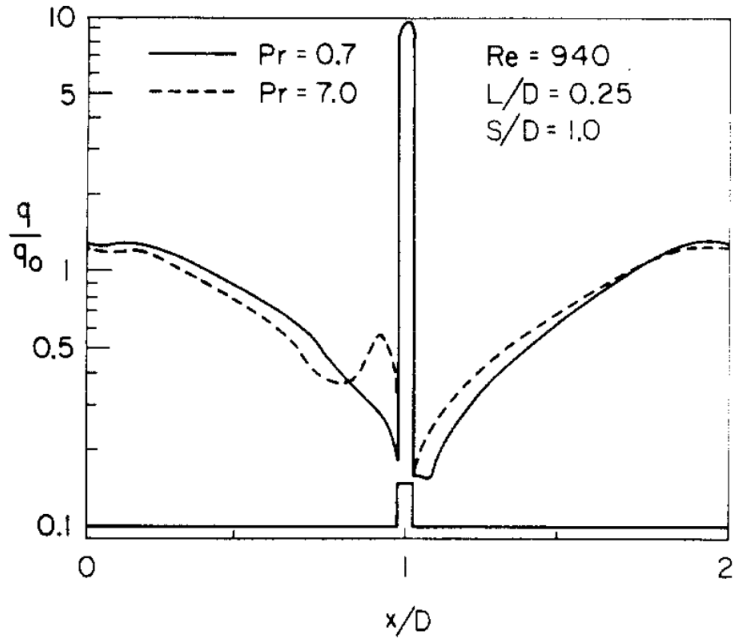


Figure 9: Local heat flux at solid-fluid interface from Webb and Ramadhyani [1].

Our corresponding results for $B/D = 0.05$ (for varying thermal conductivity ratio k_s/k_f) are shown in Figure 10. Like Webb and Ramadhyani, we see a peak in local heat flux at $x/D = 1$, or at the location of the rib. The value of the peak is lower in our case, most likely because we have a smaller L/D (the rib is smaller and doesn't protrude as far into the channel, so it does not contribute as much to the heat transfer).

We also do not observe the rise in local heat flux far from the ribs. This discrepancy is most likely due to our smaller Re. At the large Re = 940 of the figure from the reference paper, the recirculation behind the ribs is very significant and spans nearly half the channel length (Figure 12). In comparison, our recirculation region is quite small and will have a different effect on the local heat flux.

Results (cont.)

For completeness, the local heat flux for the $B/D = 0.10$ geometry is plotted in Figure 11. We see similar behavior to the thinner wall geometry. The peak at the rib is smaller because the thicker wall allows for more heat transfer in the x-direction.

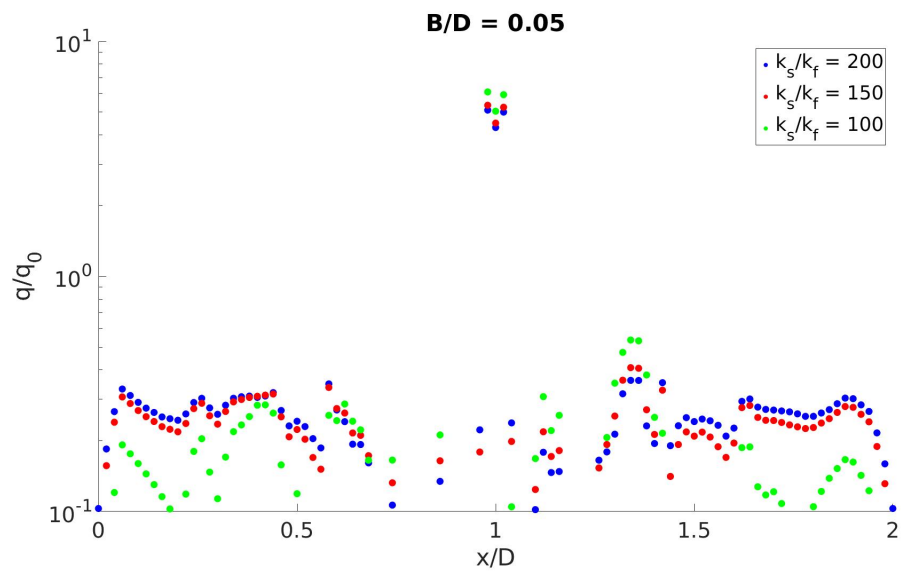


Figure 10: Local heat flux at solid-fluid interface for $B/D = 0.05$ geometry

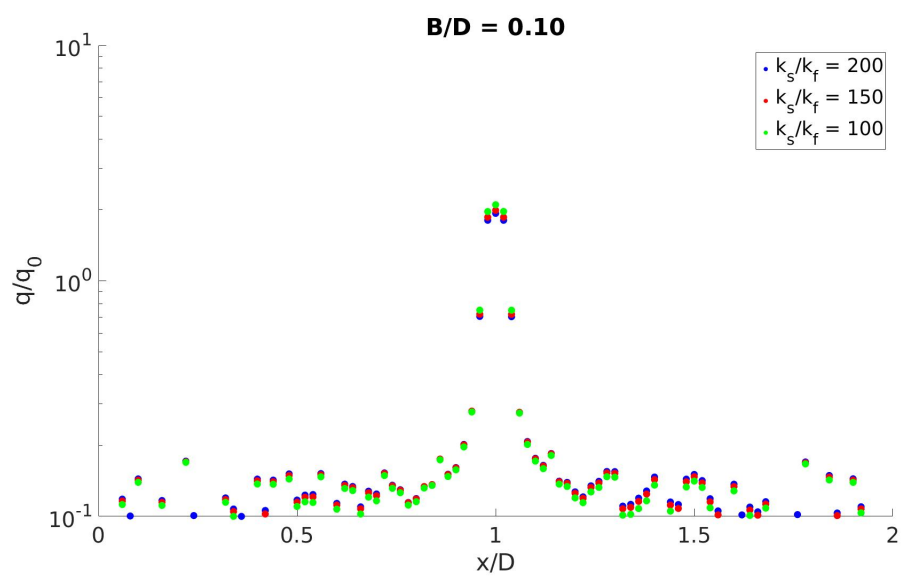


Figure 11: Local heat flux at solid-fluid interface for $B/D = 0.10$ geometry

Results (cont.)

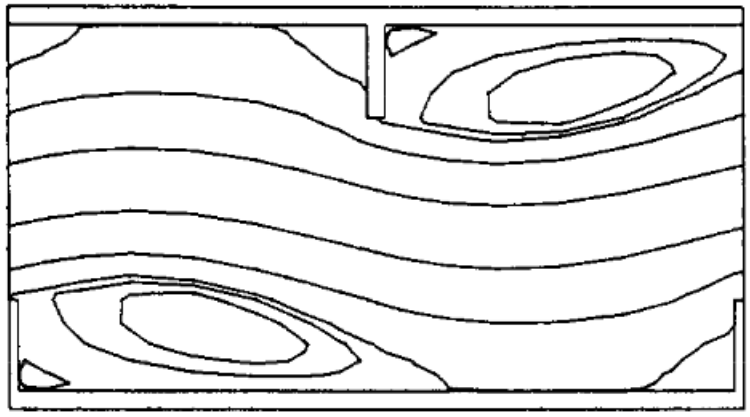


Figure 12: Streamlines of velocity field at Re = 940, $L/D = 0.25$ from Webb and Ramadhyani [1]

We now analyze the effect of varying B/D and k_s/k_f on the mean Nusselt number and on the rib heat transfer. The mean Nusselt number is defined as

$$\mathrm{Nu} = \frac{\overline{h} D_H}{k_f} = \frac{\overline{q}}{\overline{T}_{\text{surface}} - T_{\text{bulk}}} \cdot \frac{D_H}{k_f} \tag{18}$$

where

$$T_{\text{bulk}} = \frac{\int_0^{2S} \int_0^D |u| T \mathrm{d}y \mathrm{d}x}{\int_0^{2S} \int_0^D |u| \mathrm{d}y \mathrm{d}x} \tag{19}$$

and averaged quantities are averaged in x along the fluid-solid interface. The mean Nusselt number is scaled by the smooth channel value

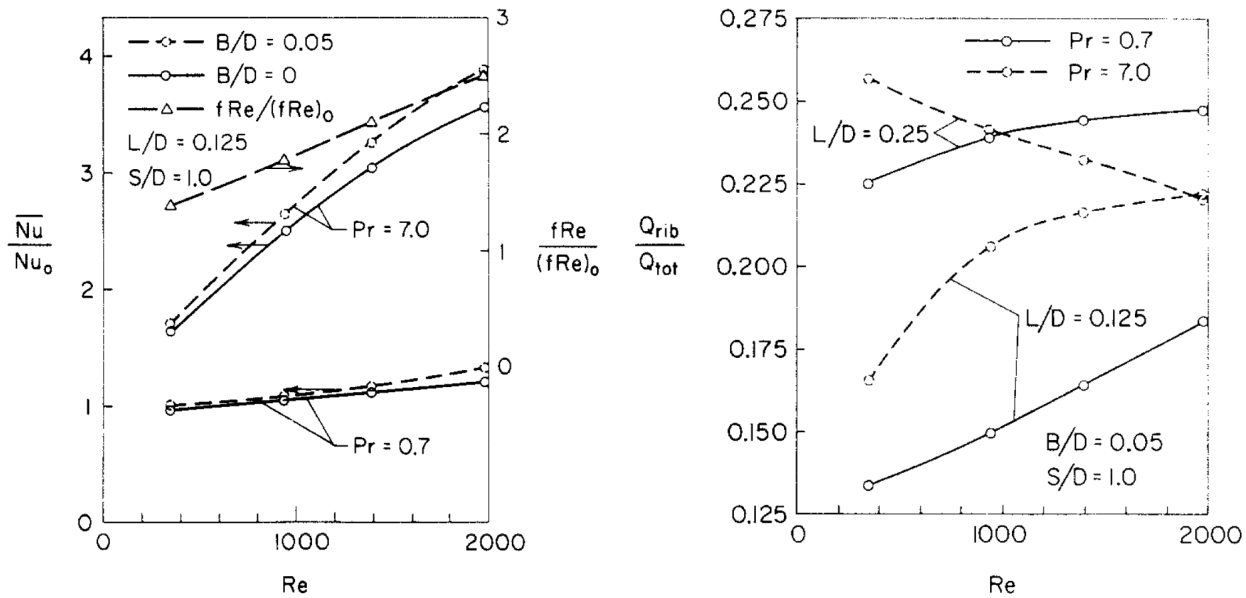
$$\mathrm{Nu}_0 = 8.23 \tag{20}$$

The rib heat transfer is defined as the fraction of the total wall heat transfer that passes through the base of the rib.

From the data presented in [1] (Figure 13, the mean Nusselt number for Re = 340, Pr = 0.7, $L/D = 0.125$, and $B/D = 0.05$ is slightly larger than 1. The rib heat transfer is approximately 0.125.

From our own simulations, we have that the mean Nusselt number is 1.108 and the rib heat transfer is 0.124. These values seem in reasonable agreement with Webb and Ramadhyani.

Results (cont.)



(a) (b)

Figure 13: Variation of (a) mean Nusselt number and (b) rib heat transfer with Reynolds number from Webb and Ramadhyani [1]

The full range of mean Nusselt number and rib heat transfer for varying B/D and k_s/k_f are plotted in Figures 14-17.

We find that generally, increasing B/D and making the wall thicker decreases the mean Nusselt number and rib heat transfer. This makes intuitive sense because the thicker wall provides a larger "capacitance" that the applied heat flux boundary condition must overcome before being exposed to the convective heat transfer of the channel flow or reaching the rib.

Decreasing the thermal conductivity ratio decreases the mean Nusselt number. This makes sense for the same reason as stated above: there is a larger "capacitance" (in this case, directly linear with thermal diffusivity α) for the heat to overcome to reach the flow field. We note that the value of $k_s/k_f = 100$ has a mean Nusselt number that is actually *less than* that for the smooth wall channel. This is not entirely surprising: we would expect that for very low thermal conductivity ratios, this will be true because the rib acts as more volume to "store" heat rather than acting as more surface area for the fluid to convect heat.

This also explains why decreasing thermal conductivity ratio increases rib heat transfer. Because the rib heat transfer is measured at the base of the rib and not at the rib surface, the increase in rib heat transfer is a symptom of this "storage" that allows for more heat to be transferred through the rib itself.

Results (cont.)

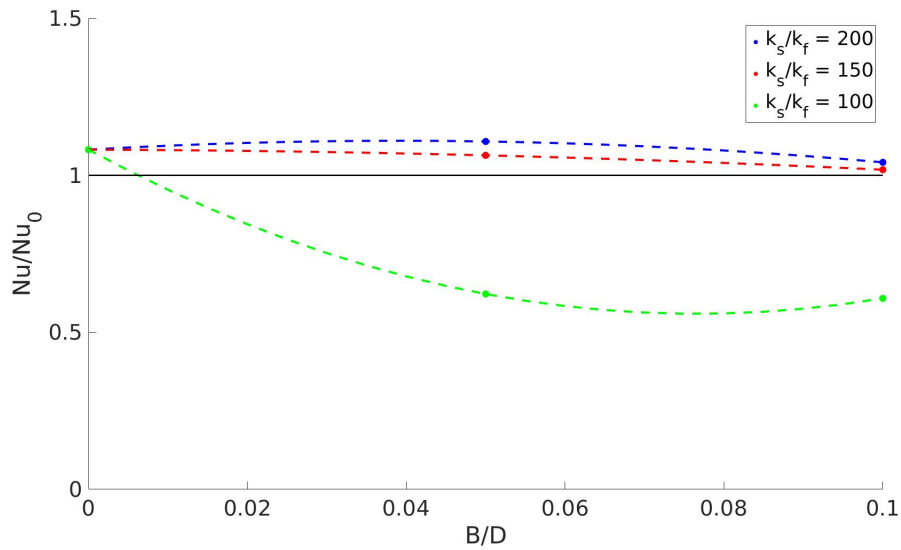


Figure 14: Variation of mean Nusselt number with wall thickness B/D

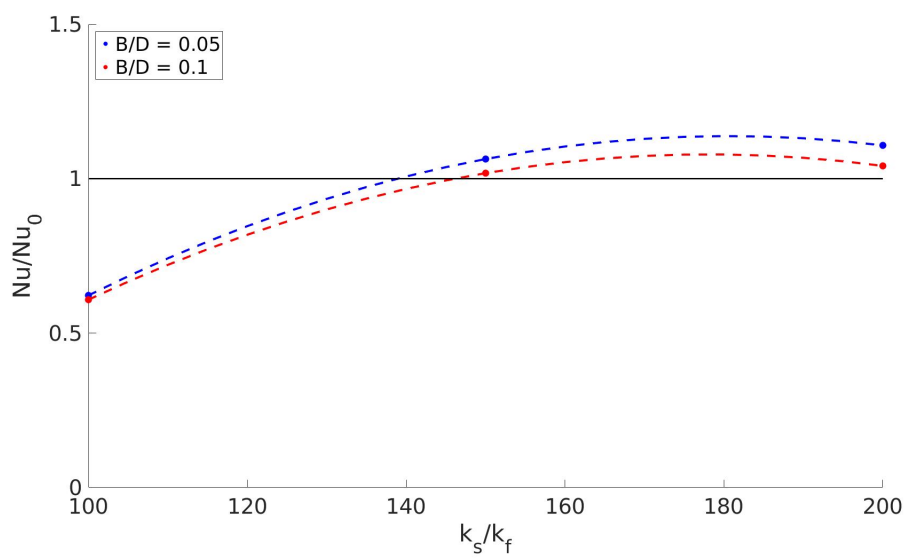


Figure 15: Variation of mean Nusselt number with thermal conductivity ratio k_s/k_f

Results (cont.)

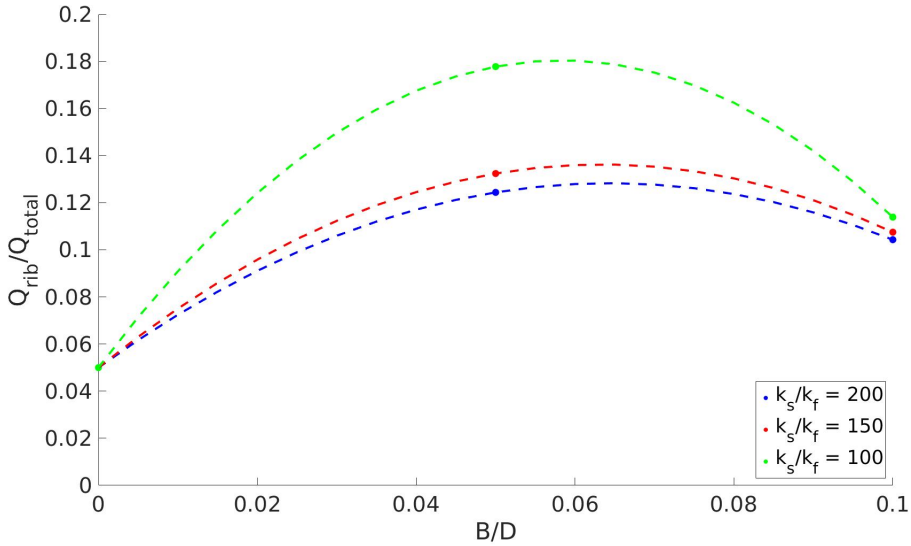


Figure 16: Variation of rib heat transfer with wall thickness B/D

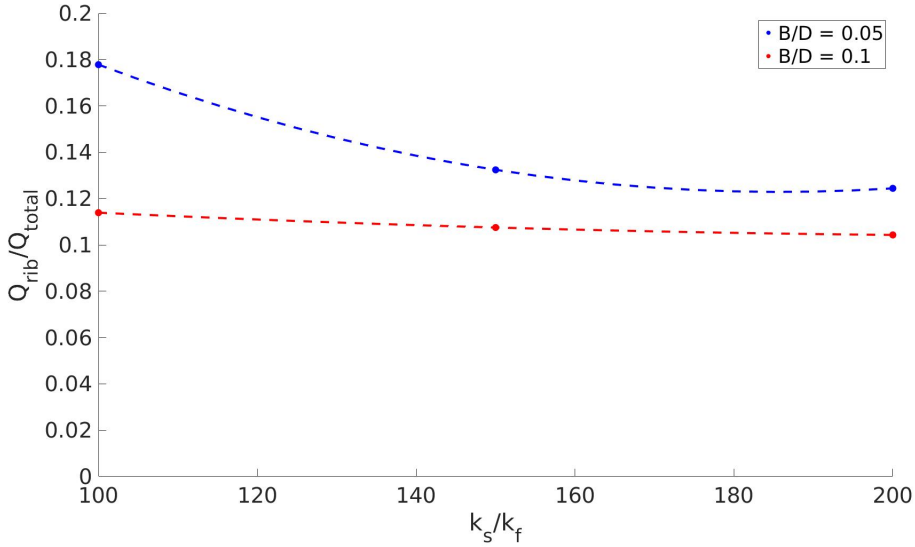


Figure 17: Variation of rib heat transfer with thermal conductivity ratio k_s/k_f

References

[1] BW Webb and S Ramadhyani. Conjugate heat transfer in a channel with staggered ribs. *International Journal of Heat and Mass Transfer*, 28(9):1679–1687, 1985.